

NANOTECHNOLOGY — SCALE & SURFACE AREA!



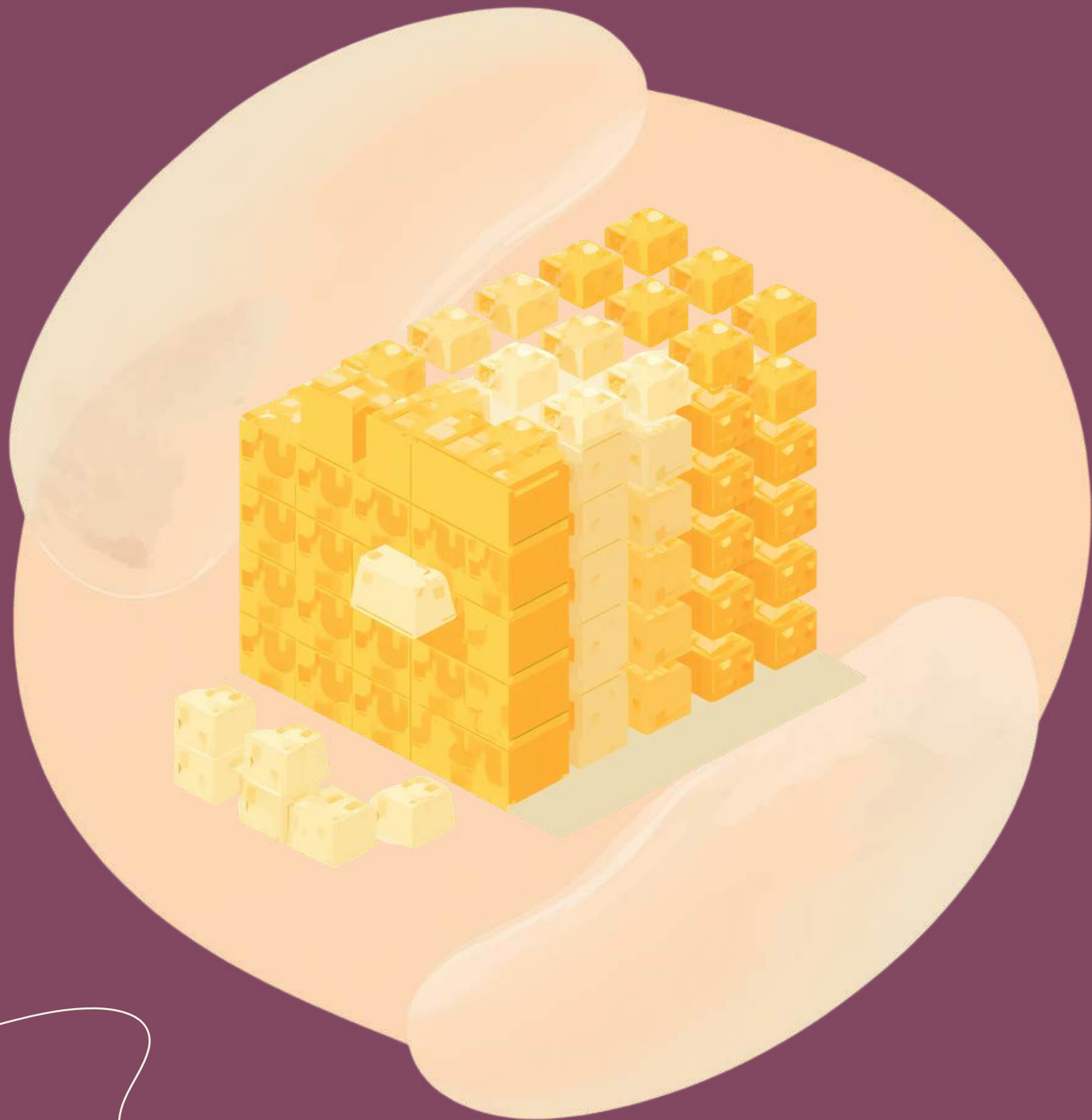


WHY DOES POWDERED GUR MELT FASTER IN WARM MILK?

Powdered gur has more surface area. More tiny pieces touch the warm milk at once, so it melts faster. A big block has less surface area exposed, so it takes longer.

EXPLORING SCALE — CRICKET BALL TO NANOMETER!

1. Cricket Ball (~7 cm) → Ant (~5 mm) → Human Hair (~80,000 nm)
2. Red Blood Cell (~7,000 nm) → Virus (~100 nm) → DNA (~2 nm)
3. Nanometer = one-billionth of a meter (10^{-9} m) — tinier than the smallest grain of sand!



**MORE PIECES =
MORE SURFACE AREA!**

- 🧩 Break one big thing into many small pieces — now more of its outside is showing!
- ✨ More exposed area touches milk, air, or water all at once.
- 🥛 That's why crushed gur, chopped paneer, or powdered tablets mix & melt super fast!



NATURE'S NANO WONDERS!

- 🌿 Lotus leaves stay clean because water rolls off tiny bumps.
- 🦎 Gecko feet help it stick to smooth walls with tiny hair-like structures.
- 💧 Nano-filters in water purifiers can catch very tiny impurities.

TRY THIS AT HOME! THE TABLET CHALLENGE!

1. Take two glasses of water.
2. Drop one whole tablet in one glass.
3. Drop crushed tablet pieces in the other.

GREAT LEARNING TODAY!
NANOMETER = SUPER TINY (10^{-9} m)!
SMALLER PIECES REACT & MELT FASTER.
NATURE USES NANO TRICKS TOO!
SEE YOU IN THE NEXT SESSION! ✨